PREPARATIONS:

- Make sure there is enough M9 (x5) media, sterile water, trace elements, MgSO4 and CaCl2.
- Make sure there are enough copies of the bacterial collection (384-well)
- Make sure there are enough autoclaved 384-well tips for Vioflow.
- Enough carbon source stocks

Day 1

1. Grow the 384-well bacterial collection:
    a. Add 40 ul NM to 384 wells transparent plates.
    b. Thaw the stock plate a little.
    c. Inoculate 5ul into the 40ul NM.
    d. Grow for 24h, shake at 1500 rpm
2. Grow PA01 in 3 ml NM.

Day 2

For 22 plates (360 ml):

- 291 ml H2O // 74 ml X5 M9 salts // 370 ul trace elements // 740 ul MgSO4 // 37 ul CaCl2 // 3.7 ml NM+carbon source of interest (0.2%) ,probably glucose
-
- Spread 40ul of the M9 media into the final amount of 384-well black LUMOX plates. To avoid bubbles, and droplets at the side of the well, use the following settings:

a. Reverse pipette with 5ul post-dispense
b. Asp speed 4. Disp speed 2-3
c. Distance - ~23.5-24 a - double check.
d. Keep at 4C covered in a bag*, (if planning to store for later)*

3. Dilute 1:5 the 24h-grown bacterial collection(*may be only the strains of your interest*) into ECHO-plate, 384-well:
    a. 32 ul PBS
    b. 8 ul bacteria

4. Dilute 1:8 the 24h-grown PA01 into ECHO-plate in PBS

5. Shake 384-well collection at **1700 rpm** for at least 3 minutes **& remove bubbles manually (can also centrifuge).**

6. Use the picklist to shoot the combinations

Remember to shake the source plate in between.
Ensure plate reader is not running while using the echo
Test shoot if possible,ensure the droplet somewhat goes to the center
There are some minor dimension changes in greiner plates among batches ,ensure using the right plate is selected ,also greentape the plates to ensure echo grips well

7. Add PBS + 1:10 NM as control into the ECHO-plate
   *Make sure to keep the pathogen to commensal ratio to 1
8. Shoot 100 ul pathogen PA01 and negative control (PBS+NM 1:10)
9. Place the plates at 30 C, **inside a bag with water for high humidity.**

Check the run status of the echo to ensure the protocol had no exceptions

Day3

10. Fill in Lumox plates with the 40 µl per well media
11. Add the culture 1µl (change according to dilution ratio)
12. KeeP them covered in 30 C

For measurements,

13. Add 9 ul of NM to plates. Use the following Vioflow settings:

14. **Place the plates in a dark box (the orange one). Keep it dark between measurements.**

15. **Wait 2 minutes before measuring the first plate.**
16. For these 5 plates: measure only luminescence using the fast readout protocol - **24 h after the first bacteria was shot. Change the file name of every measurement to the right one.**

17. **In between - shake the next plate at 1300 rpm for 45 seconds.**

Careful if using the longer protocol with the plate reader!

Make sure to keep the time difference between the plates to be under 15 min,from adding nm in the first plate to measuring lum last plate.

May be it's worth trying to plate the source plate ,and some random communities to see if there is any contamination in random days,also maybe it would give some info on community composition ,stability?

SInce I used low volume plates ,I did 20 culture + 4.5 µl NM ,used xiazhou's protocol ,head and did integration time 2 in this case ,for the white lumox plates with no well wall sharing

*probably should stick to this protocol and plates,and sensitivity head as the results with the model was better .

Picklist strategies

Ensure there is an equal number of samples for all communities?

Ensure all plates have negative and pathogen only controls ,use this to normalise between plates ?

Also keep replicates on all plates too.

Ensure these controls and wells are randomised ,while keeping the community size distribution similar in all plates to ensure they spend the same time in the echo .

For ss plate ,have at least three replicates of each strain ,have negative and pathogen controls in the plates too..randomise the well order.